Contrastive Feature Disentangling for Partial Aspect Angles SAR Noncooperative Target Recognition

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***Abstract*— Deep learning algorithms have achieved state-of-the-art progress in synthetic aperture radar (SAR) automatic target recognition (ATR) tasks. They theoretically assume that training and test samples are independent and identically distributed (i.i.d.) for generalization, but it is intractable for practical ATR scenarios. In this article, we propose a novel contrastive feature disentangling framework termed ConFeDent to learn features with improved generalization performance under a condition of a weaker distribution consistency. More specifically, ConFeDent aims to describe the semantic interactions between two arbitrary SAR training samples instead of treating them independently. It can implicitly disentangle features encoding the pose and identity knowledge from the whole samples with a semi-parametric geometric transformation model and a second-order energy model. In particular, except for the identity label, we use deductive-based geometry knowledge as supervision to teach the model to learn the concept of aspect angle variation. A progressively amortized inference scheme is constructed for efficient feature learning and recognition in an end-to-end manner. Finally, we further release a strengthened version, called ConFeDent+, which can explicitly use and learn more information from cross-category samples. Experimental results on the moving and stationary target acquisition and recognition (MSTAR) benchmark demonstrate the effectiveness of our proposed models in the SAR ATR. In particular, we validate the algorithms in a more challenging scenario where the range of aspect angles for training and testing samples is permitted to be disparate. Our model can achieve much higher recognition accuracy than other SAR ATR algorithms.**

***Index Terms*— Automatic target recognition (ATR), contrastive learning, extended operating condition (EOC), feature disentangling, out of distribution (o.o.d), partial aspect angles (PAAs), synthetic aperture radar (SAR).**

I. Introduction

AUTOMATIC target recognition (ATR) is one of the ultimate goals in the field of synthetic aperture radar (SAR), which has attracted much attention in civil and military reconnaissance and surveillance for years [1], [2], [3]. Unlike optical sensors, the electromagnetic imaging mechanism of SAR makes the resulting image reconstruction from the specular backscatter of the illuminated target [4]. Therefore, SAR will actually "see" some types of physical structures of a target, and scattering signatures in the resulting SAR image will be highly sensitive to geometric acquisition conditions, e.g., azimuth, and depression angle. Crucially, the profile and the topological structure of scattering points of the target in the image domain will vary greatly as the viewing angle of the SAR platform changes. The issue of geometric acquisition sensitivity will make SAR ATR more challenging than a general image recognition task of optical sensors.

To address this problem, conventional scattering center model-based SAR ATR algorithms collect the configurations of strong scattering points/centers of the target in full (0–2π) aspect angles [5], [6]. These complete scattering patterns form a feature template in the hope of uniformly collecting all the scattering signatures in different poses. Pairwise template matching is then used to classify a test target. However, the scattering features are affected by many practical disturbances, such as speckle noise and motion ambiguity, which degrade the performance. To improve the robustness, several hand-crafted visual feature extractors are designed to capture some pose-invariant signatures of the target image. These symbolic visual features are more robust to pose variations to achieve better performance than scattering features [7]. A classifier is learned in the precomputed feature space. The paradigm of deductive feature extraction followed by classifier learning has dominated for years. With the increase in available computational power and sufficient samples, the learning process has shifted from the hand-crafted feature domain to the raw sensory data domain, allowing us to learn features and a classifier simultaneously. Accordingly, many machine-learning-based SAR ATR algorithms have been developed [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], among which the most notable is the convolutional neural network (CNN). Unlike the deductive feature extraction algorithms, the learning model addresses the pose sensitivity problem inductively. It aims to find a parametric function from a set of rotating targets and their class labels to implicitly induce rotation-invariant features. This end-to-end learning pipeline can significantly improve the accuracy and efficiency of ATR.

A. Motivation

Notwithstanding the undeniable success of the above inductive learning models, especially CNN, critics have recently drawn attention to a number of following limitations.

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1) *Weak Generalization Ability:* Compared with the conventional deductive features, the generalization ability of the inductive learning model is based on the theoretical assumption that the training and test samples are statistically drawn from an identical distribution. However, due to the sensitivity of the SAR image to the geometric acquisition condition [2], [14], [19], the practical acquisition condition of a test target may be different from the training one, including variations in depression angle, resolution, target configuration, environmental electromagnetic noise, and so on. It is intractable to collect a large volume of targets to cover enough patterns to ensure distribution consistency, resulting in the so-called small sample problem [21]. In particular, for a noncooperative (NC) target, we can only reconnoiter a few samples from limited viewing angles for training, while test scenarios are unlimited. In this scenario, it is prone to fail catastrophically once exposed to samples outside the training distribution, leading to the complicated out-of-distribution (o.o.d.) classification problems [22].
2) *Low Interpretation Ability:* The current learning model will mostly induce the concept from the samples in the spirit of mathematics rather than the human mind. As a result, it is typically a "black box induction" process where the computations are distinct from human comprehensive reasoning and inference. The resulting intermediate features are generally semantically meaningless. As such, the learning-based SAR ATR algorithms may involve some potentially incalculable risks.

The above two limitations raise an urgent need to develop a learning-based algorithm with improved generalization and interpretation capabilities for practical SAR ATR. A simple and plausible way to alleviate the o.o.d. problem in computer vision would be a data augmentation trick or novel view generation [23], i.e., manually generating pseudotargets with full aspect angles for distribution completion and alignment. However, we have empirically found that this trick is invalid for SAR images. It may be due to ambiguous artifacts such as over-smoothness caused by nonlinear pixel interpolation, which does not correctly reflect the actual physical scattering signature of the real target. Therefore, the pseudosamples do not explicitly provide more discrimination information than the original ones [18], [24], [25]. Alternatively, computer simulation technologies, such as Microwave Studio Asymptotic Solver, can be used to generate the simulated targets for training and transfer the full aspect angle information to the real data domain via a transfer learning algorithm [4], [26]. However, this scheme relies on an accurate 3-D computer-aided design target model [4], which is intractable for NC targets. The fundamental issue of improving the generalization and interpretation of induced features is not addressed. Wen et al. [21] proposed a rotation-aware framework that aims to capture the rotation equivariant and label-invariant features of the SAR target. They used a self-supervised strategy to force the model to understand the rotation variance among multiple intraclass samples, so that it can mitigate the influence of rotation sensitivity and improve the classification performance under limited training samples. However, the framework does not consider the challenging partial aspect angle (PAA) situation. In addition, the interpretability of the resulting rotation equivariant features is also not guaranteed.

B. Contribution

This article proposes a novel contrastive feature disentangling framework, called ConFeDent, which aims to alleviate the above problems. Instead of learning the pose-invariant features for each target independently in a discriminative manner, ConFeDent relies on disentangling the pose features from their identity features by exploiting the semantic equivariant relationship between two targets in a contrastive manner. In this way, it can implicitly learn and transfer the complete pose knowledge from the cooperative training samples to the NC ones and improve the generalization ability and interpretability of the learned features for SAR ATR. To this end, ConFeDent elaborates a semi-parametric geometric transformation model and a second-order energy model as two semantically explainable regularizers to capture the structured and contrastive relationship in the feature space. Furthermore, ConFeDent incorporates a progressively amortized inference for efficient feature extraction and recognition via an encoder–decoder scheme. Finally, we release a strengthened version termed ConFeDent+. The experimental results on the moving and stationary target acquisition and recognition (MSTAR) benchmark demonstrate the effectiveness of our proposed approach. Compared with the other ATR algorithms, both ConFeDent and ConFeDent+ can achieve higher recognition accuracy in the SAR ATR task, especially in the PAA ATR. The main contributions can be summarized as follows.

1) We propose a novel contrastive feature disentangling model, called ConFeDent, and an enhanced version, ConFeDent+, to induce features with improved generalization and interpretation ability. Compared with the other SAR ATR algorithms, the proposed models can achieve much better SAR ATR performance not only under the general standard operating condition (SOC) and extended operating conditions (EOCs) but also under a more complicated PAA scenario.
2) We develop a novel idea of learning to disentangle the pose and identity features by comparing the semantic relations between two arbitrary SAR targets instead of treating them independently. The induced pose feature subspace of all the training samples will contain complete aspect angle information. The missing aspect angle information of the NC samples will be sampled and transferred from those in cooperative/aspect available classes, which essentially mimics the process of human imagination to increase the combinational generalization ability.
3) We incorporate a semi-parametric geometric transformation model and a second-order energy model for contrastive regularization. In addition to using the identity label for individual supervision, we specifically elaborate the deduction-based geometry knowledge as pairwise relation supervision to teach the model to learn the concept of relative aspect angle. The novel insight of

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using the relative relation as supervision instead of the absolute one will shed new light on the development of a new learning strategy. By taking the best of both worlds, it inherits the advantages of both deductive and inductive learning, analogous to the human way of learning.

The rest of this article is organized as follows: Section II introduces the preliminary. Section III formally proposes the main framework of ConFeDent and its detailed architecture. Extensive experiments are carried out in Section IV to demonstrate the effectiveness and superiority of the proposed model. Section V summarizes our work and suggests future directions.

II. Problem Formulation and Analysis

The aims in this section are threefold. First, we formally propose the new type of EOC to formulate the problem of PAA ATR and analyze some insights from the perspective of SAR imaging model. Second, we discuss the equivariant property of features to shed light on our solution to PAA ATR. Finally, we introduce the related research on contrastive learning to present our idea.

A. EOC of PAA and Insight

Considering the spotlight mode SAR imaging model [27], the received raw data of an illuminated ground target from viewing angle θ_c and depression angle φ can be roughly described by $\mathbf{r}_{\theta_c,\varphi}(\omega_c) \leftarrow \mathcal{A}_{\theta_c,\varphi}(x)$, where ω_c is the radar center frequency, $\mathcal{A}_{\theta_c,\varphi}$ is the radar measurement operator, and x is the ontology-based target conditioned on its category $\mathcal{C}$ and pose (mainly yaw angle). The phase history matrix $\mathbf{R}_\varphi(\omega_c, \theta_c) = [\mathbf{r}_{\theta_1,\varphi}(\omega_c), \ldots, \mathbf{r}_{\theta_K,\varphi}(\omega_c)]$ is composed of a number of signals with continuous viewing angles $\{\theta_c \pm k\Delta\theta\}, k = 1, \ldots, K/2$ with interval $\Delta\theta$, and K counts the number of viewing angles. It follows that the SAR image of x can be obtained from $\mathbf{R}_\varphi(\omega_c, \theta_c)$ with an appropriate imaging algorithm $\mathcal{I}$ as $\mathbf{x} = \mathcal{I}(\mathbf{R}_\varphi(\omega_c, \theta_c))$. According to this model, we can conclude that the SAR image $\mathbf{x}$ is causally controlled by many factors, including ω_c, θ_c, φ, and x. From the perspective of the probability model, $\mathbf{x}$ is a sample drawn from $P(\mathbf{x}|\omega_c, \theta_c, \varphi, \mathcal{C})$ where the yaw angle of x can be absorbed in θ_c. According to this model, the task of ATR is to infer the posteriori distribution of $\mathcal{C}$ according to $\mathbf{x}$ and other available factors. Due to the electromagnetic imaging mechanism, the appearance of $\mathbf{x}$ will be highly sensitive to θ_c and φ. A slight pose variation in x will yield an apparent change in $\mathbf{x}$. The conventional algorithms to alleviate this challenge will exploit some physically or geometrically driven methods to extract some rotation-invariant features in a deductive way. Alternatively, the inductive-based learning model aims to adaptively induce invariant features from training samples drawn from the marginal distribution $P(\mathbf{x}|\omega_c, \mathcal{C}) = \iint P(\mathbf{x}|\omega_c, \theta_c, \varphi, \mathcal{C})P(\theta_c)P(\varphi)\mathrm{d}\theta_c\mathrm{d}\varphi$ according to ancestor sampling. With this marginal distribution, the influence of θ_c can be avoided when inferring $\mathcal{C}$ from $\mathbf{x}$. Compared with conventional hand-crafted feature engineering, the inductive learning-based ATR model can discover new data-adaptive solutions from these training samples and probably create new recognition knowledge beyond those canonical ones. However, it is not applicable in a realistic battlefield deployment scenario of SAR ATR since $P(\theta_c)$ and $P(\varphi)$ will generally be different from the training phase. Concerning $P(\theta_c)$ as an example in this article, we can only reconnoiter a few NC samples with limited poses for training, where $\theta_c \sim P(\theta_c)$ can only be drawn from a uniform distribution $\mathcal{U}(\theta_l, \theta_u)$, $[\theta_l, \theta_u] \subset [0, 2\pi]$ while θ_c in the testing phase may be drawn from $\mathcal{U}(0, 2\pi)$. The θ_c of the training samples in this case will only reside in a partial range such as $[0, \pi]$. Although $P(\mathbf{x}|\omega_c, \theta_c, \varphi, \mathcal{C})$, which reflects the physical imaging mechanism, will be invariant under the causal structure, $P(\mathbf{x}|\omega_c, \mathcal{C})$ will change in a so-called localized way [28]. In the causal learning area, it is known as the independent causal mechanism principle with the sparse distribution shift. In this case, the underlying test distribution will be different from the training distribution, yielding the o.o.d. learning and recognition problem. It is defined as the new EOC termed PAA for SAR ATR, which requires more generalization ability to recognize unseen views. PAA can also be served as a complementary condition in conjunction with the conventional EOCs in Table I [2], which form much more challenging verification conditions. It should distinguish the PAA condition from the problem of missing azimuth data in the phase history for SAR imaging [29].

TABLE I
Summary of Current EOCs [2]

extended operating conditions	specification
acquisition geometry	depression angles squint angles
target state variations	articulation alternative configurations intra-class variability
local target deployment	obscuration: occlusion and layover camouflage deception

B. Equivariant Feature Disentanglement

The deep-learning-based SAR ATR algorithms classify a query sample $\hat{\mathbf{x}}$ according to the following optimization:

$$\arg\max_c \mathcal{E}^*(\hat{\mathbf{x}}), \quad \text{s.t. } \mathcal{E}^* \in \arg\min_{\mathcal{E}} \sum_c \sum_{i=1}^{N_c} \ell(\mathbf{y}_i, \mathcal{E}(\mathbf{x}_i)) \tag{1}$$

where ℓ is a classification loss function, $\mathcal{E}$ is a designed discriminative model such as a CNN, $\mathbf{y}_i$ is the one-hot label vector and N_c is the number of training samples in the cth class. If $\hat{\mathbf{x}}$ and $\mathbf{x}_i$ are drawn from the same distribution, (1) can theoretically learn a well-generalized $\mathcal{E}^*$ that achieves satisfactory classification performance. Otherwise, the performance will be degraded because the canonical CNN has little ability to imagine and learn the features representing unseen samples o.o.d. Let $\mathbf{x}_i$ and $\mathbf{x}_j$ be two input SAR target images from the same class. Assuming that there is an oracle function $\mathcal{T}_{i\to j}$ accounting for the transformation from $\mathbf{x}_i$ to $\mathbf{x}_j$, it will entirely capture their similarity and dissimilarity. Then there is a group of $\mathcal{T}_c = \{\mathcal{T}_{i\to j}\}_{i,j=1,\ldots,N_c}$ encoding the intraclass variations among the samples, e.g., variation in θ_c. Note from (1) that the discriminative learning will facilitate a function

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invariant to these transformations as

$$\mathcal{E}^*(\mathbf{x}_i) \approx \mathcal{E}^*(\mathbf{x}_j) = \mathcal{E}^*(\mathcal{T}_{i\to j}(\mathbf{x}_i)) \quad \forall i, j. \tag{2}$$

Therefore, if the training samples can implicitly cover the underlying $\mathcal{T}_c$ or are explicitly augmented by $\mathcal{T}_c$, $\mathcal{E}^*$ will be invariant to $\mathcal{T}_c$ conditioned on its hypothesis space. Since canonical CNN does not involve an explicit spatial rotation operator, models in this hypothesis space can only produce local rotation-invariant features through a deep hierarchy of pooling layers. Laptev et al. [30] improved the rotation invariance of CNN by proposing a transformation-invariant pooling operation. Jaderberg et al. [31] proposed a spatial transformation network (STN) module to transform the intermediate features in a discriminative manner. Cohen and Welling [32] leveraged a smaller symmetric group transformation including a flip and four 90° rotations. The invariant features can be approximated by pooling within the group. Nevertheless, the above approaches are inappropriate for PAA ATR.

Alternatively, in addition to learning invariant features with the discriminative model, the research of discovering equivariant features with generative learning has also attracted increasing attention [33]. It assumes that the input sample is latently controlled and generated by some independent factors. The variations among samples are essentially caused by their change, which is given by

$$\mathbf{x}_j = \mathcal{G}^*(\mathbf{z}_j) = \mathcal{T}_{i\to j}(\mathbf{x}_i) = \mathcal{T}_{i\to j}(\mathcal{G}^*(\mathbf{z}_i)) = \mathcal{G}^*(\hat{\mathcal{T}}_{i\to j}(\mathbf{z}_i)) \tag{3}$$

where $\mathcal{G}^*$ represents the underlying data generation process entangling the semantically explanatory factors/features $\mathbf{z}$, and $\hat{\mathcal{T}}_{i\to j}$ is a corresponding transformation in the latent space. By extracting and disentangling the explanatory factors from $\mathbf{x}$, it is easier to recover more information about the input domain. The factors can generally contribute to downstream tasks and make the model transparent [33]. This article indicates that the latent target pose features are globally shared across all the categories. Although the viewing angles of the targets in the NC categories are incomplete in PAA, the missing information can be imagined and transferred from the samples in the cooperative categories. To this end, our solution for PAA ATR is to develop a feature disentanglement model to separate the class-shared pose factors from the class-specific identity ones and to transfer the complete viewing angle knowledge from the cooperative classes to the NC.

Nevertheless, the objective function of a generative model for feature disentangling is more expensive to evaluate and more difficult to design than (1) since it focuses on the latent domain without explicit ground truth, resulting in an ill-posed inverse problem. Consequently, prior- or bias-induced constraints or regularizations are necessary to recover the underlying desired features [34], [35], [36], [37], [38]. Overall, two types of strategies have been devoted to feature disentangling, namely, variational-Bayes inference [39] and adversarial learning [40]. InfoGAN exploits an information-theoretic regularization in the generative adversarial network (GAN) for disentangling [41]. β-variational autoencoder (VAE) [39] overpenalizes the posterior distribution of the factors in the VAE [42] to push it toward a fully factorized prior distribution for disentangling. To further improve the performance of β-VAE, the total correlation was used as a regularization in the factor VAE [36] and the β-TCVAE [43]. Other metrics such as covariance are also used for learning disentangled representations [44]. Nevertheless, the above models still treat each training sample independently, and thus they cannot explicitly capture the semantic interactions among multiple samples such as rotation and identity. It is difficult for the above models to disentangle the factors that account for the desired semantic information without additional supervision.

C. Contrastive Learning

Contrastive learning is attractive because of its success in unsupervised representation learning [33], [45]. As a branch of self-supervised learning, it learns representations based on a pretask of comparing the similarity and dissimilarity between samples. More specifically, instead of treating each training sample individually, it learns to compare two inputs to extract the representations that capture similar and dissimilar information in terms of the specified semantic criterion, such as category. Then, the inductive bias that the representation of positive samples should be close while that of negative samples should be far away will guide and regularize the learning process without additional supervision. It will not only alleviate the labeling limitation encountered in supervised learning but also provide a scalable and accessible mechanism to specify the desired invariant or equivariant properties of the learned function [46]. In the contrastive learning paradigm, an encoder is first learned to map two input samples to representations in a latent space. Although the dimension of the latent space is smaller than the input domain, it always requires a large computational effort to measure the similarity. To alleviate this problem, a projection head will further project the representations into an embedding space with a much lower dimensional space equipped with a contrastive loss function for measurement. More detailed information can refer to [46] and references therein.

As mentioned above, measuring similarity or dissimilarity with respect to a semantic concept plays a crucial role in contrastive feature learning. It provides an inductive bias to encode the desired transformation equivariant property into the learned features without manual labeling effort. Inspired by this advantage, this article will develop a contrastive feature disentanglement framework by designing the metrics in terms of pose and identity similarity between two SAR images.

III. Proposed Framework

In this section, we will first develop the contrastive feature disentangling model. Then, we develop two contrastive regularizations on the identity and pose factors, respectively. Next, we will propose a regularized encoder–decoder architecture for amortized inference and model learning. Finally, a strengthened version termed ConFeDent+ is developed.

A. Contrastive Feature Disentangling Model

Note from the discriminative learning objective function (1) that the external knowledge used to teach the machine model to classify a target is its image and identity label. However,

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the image conceals abstract semantics such as the target pose, while the label, symbolized as a one-hot vector, reflects neither the individual characteristics of intraclass variances and interclass similarities nor the semantic concept of this class. In this sense, if the pose identity patterns of an NC category are highly incomplete in the training set, it is intractable for the above inductive learning scheme to generalize the classification ability to an NC target with an unseen view angle. Alternatively, a human can generalize this ability from the perspective of semantic perception and imagination. Moreover, when considering two target images from different categories, the human visual cognition system can also measure the pose similarity of the targets in an implicit disentangling manner. These observations suggest that patterns contain more than identity and that much more representative information is hidden in the entire training set. Therefore, the main idea of this article is to learn from populations, i.e., to use the semantic relationship between populations as an inductive bias, which leads to a new feature learning paradigm with improved generalization. It will be a new solution to the PAA ATR task by sharing information across categories. The complete knowledge of aspect angles will be extracted and exploited by transferring from cooperative classes to NC classes.

Let $\mathbf{r}$ and $\mathbf{f}$ be the latent target pose (relative to the radar) and identity (class-specific) factors that account for θ_c and $\mathcal{C}$, respectively. Based on the imaging mechanism in Section II-A, the SAR target image $\mathbf{x}$ of a specific radar is partially controlled by $\{\mathbf{f}, \mathbf{r}\}$ through a nonlinear parametric function as $\mathbf{x} \leftarrow \mathcal{G}_\Theta(\mathbf{f}, \mathbf{r})$, where Θ contains the parameters. The $\{\mathbf{f}, \mathbf{r}\}$ of all the training samples will be in two disjoint semantic subspaces. For the identity subspace $\mathcal{F}$, it can be characterized as a normal Euclidean space. For the pose subspace $\mathcal{R}$, since there are always cooperative targets in the training set, the entire $\{\mathbf{r}_i\}_{i=1}^{N_c}$ can cover this subspace as densely as possible. They can encode the complete aspect angle information of $[0, 2\pi]$ even if they are not evenly distributed in the PAA condition. Accordingly, $\mathcal{R}$ can be modeled as a manifold. Note that the union of the subspaces $\mathcal{F} \bigcup \mathcal{R}$ essentially forms a feature knowledge pool in which all the disentangled information is shared for combinational generalization [38]. In fact, the human brain can arbitrarily sample from the knowledge pool to imagine and create new concepts by (reasonable) combination, even though most of them may not exist in reality. It makes the brain creative and generalizable. Accordingly, the manner of decomposition–composition in the above model is analogous to the knowledge processing and generalization of the human brain. To disentangle the above factors, we intend to teach the model to understand two target images by comparing their similarities and differences in identity and pose, respectively. For two targets $\mathbf{x}_i$ and $\mathbf{x}_j$, the contrastive feature disentangling framework can be initially formulated as

$$\begin{aligned} \min_{\{\mathbf{r}_i, \mathbf{f}_i\}_{i=1}^{N_c}, \Theta} & \sum_{c=1}^{C} \Omega_f\left(\{\mathbf{f}_i\}_{i=1}^{N_c}\right) + \alpha \sum_{c=1}^{C} \Omega_r\left(\{\mathbf{r}_i\}_{i=1}^{N_c}\right) \\ \text{s.t. } \forall i, \quad & \mathbf{x}_i \leftarrow \mathcal{G}_\Theta(\mathbf{f}_i, \mathbf{r}_i), \ \ \mathcal{F} \perp \mathcal{R} \end{aligned} \tag{4}$$

where $\Omega_f(\cdot)$ and $\Omega_r(\cdot)$ are two contrastive regularization functions, and α is a hyperparameter for balance. Note from (4) that it originates from the idea of independent component analysis (ICA) for source separation [47] and robust principal component analysis (RPCA) [48]. As aforementioned, the design of $\Omega_f(\cdot)$ and $\Omega_r(\cdot)$ will play an important role in equivariant feature disentanglement, which will be addressed in Sections III-B and III-C.

B. Contrastive Identity Factor Regularization

The first regularizer $\Omega_f(\cdot)$ measures the similarity of class-specific factors related to the identity. To characterize this metric, we resort to the second-order energy model for pairwise relational learning [49]. More specifically, $\{\mathbf{f}_i\}_{i=1}^{N_c}$ form a pairwise random field conditioned on their identity labels, and $\Omega_f(\cdot)$ is designed to measure its potential energy

$$\Omega_f\left(\{\mathbf{f}_i\}_{i=1}^{N_c}\right) \equiv \sum_i \omega_u(\mathbf{f}_i; \mathbf{y}_i) + \mu \sum_i \sum_j \omega_p(\mathbf{f}_i, \mathbf{f}_j; \mathbf{y}_i, \mathbf{y}_j) \tag{5}$$

where ω_u and ω_p are the unary and pairwise potential energy functions, respectively, and μ is a balance parameter. Considering first the unary potential ω_u, it measures the compatibility between a supervised label $\mathbf{y}$ and the identity feature vector $\mathbf{f}$. An intuitive way is to learn a projection head $\mathcal{W}(\cdot)$ to embed $\mathbf{f}$ in the label space, where the similarity can be derived from the cross-entropy loss function as $\omega_u(\mathbf{f}; \mathbf{y}) = \text{CE}[\mathbf{y}, \mathcal{W}(\mathbf{f})]$. Note that $\mathcal{W}(\cdot)$ should be as concise as possible to encode enough class-specific information in $\mathbf{f}$. An over-parameterized $\mathcal{W}(\cdot)$ can warp a simple noninformative distribution into a complex one, in which case the regularization becomes invalid. Accordingly, in our model, it is designed as a one-layer perceptron with a softmax activator. For the pairwise potential ω_p, it simply measures the similarity between $\mathbf{f}_i$ and $\mathbf{f}_j$. The properties of positive pairs drawn from the same class $\mathcal{C}^+$ should be similar, while otherwise, they should be different. When the dimension of $\mathcal{F}$ is high, a projection head $\mathcal{P}$ is needed to embed the features in a lower dimensional space equipped with a contrastive metric. Three forms of loss functions have been used in current contrast learning models [46], including triplet loss, probabilistic noise contrastive estimation-based loss, and distance functions. We empirically found that the Euclidean or Manhattan distance can achieve improved performance on the validation set. As a result, we consider the following function:

$$\omega_p(\mathbf{f}_i, \mathbf{f}_j; \mathbf{y}_i, \mathbf{y}_j) = \begin{cases} \|\mathcal{P}(\mathbf{f}_i), \mathcal{P}(\mathbf{f}_j)\|_q^q, & \text{if } \mathbf{y}_i = \mathbf{y}_j \\ \text{const}, & \text{otherwise.} \end{cases} \tag{6}$$

So far, we have almost finished designing the regularization function for the discriminative factors. The unary and pairwise potentials will impose different regularizations on $\mathbf{f}$. More formally, ω_u allows the model to capture identity information, while ω_p promotes consistency between two intraclass vectors. Nevertheless, we find that ω_p turns to over-punishment in the early training phase, leading to a catastrophic collapse. In this case, ω_p quickly reaches its lower bound by forcing two vectors to be identical vectors for all the inputs. It is reasonable to introduce a delayed mechanism before the model has extracted discriminative factors. In our framework, we use a progressive regularization scheme by dynamically varying the

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value of $\mu = (1 - \exp(-10\rho))/(1 + \exp(-10\rho))$ from 0 to 1, where ρ is the ratio of the current epoch to the maximum steps of the iterations.

C. Pose Contrastive Transformer-Based Regularization

For the second regularizer $\Omega_r(\cdot)$, it characterizes the pose factors and measures their similarity on a manifold $\mathcal{R}$. Compared with Ω_f, new challenges arise from the following two perspectives. First, since it is intractable to explicitly and accurately annotate the entire continuous manifold, we cannot construct a supervised unary function to encode the pose information in $\mathbf{r}$ as ω_u does. Second, since $\mathcal{R}$ is a continuous manifold, the general Euclidean metric such as ℓ_1 of ℓ_2-norm is not suitable to measure the similarity of points in it.

To address the first challenge, we specifically exploit the deductive-based geometry knowledge as supervision to teach the model to learn the concept of "what is the pose of a target." Instead of using the absolute variable of θ_c as supervision to regularize $\mathbf{r}$, we innovatively characterize it with the relative value by comparing it with the other targets. More specifically, considering the pose factor $\mathbf{r}_i$, if it really captures the whole pose information of $\mathbf{x}_i$, it can be described by the factors of others since there will be an explicit operator $\mathcal{T}_{\theta_{j\to i}}, \forall j \neq i$ warping $\mathbf{r}_j$ to $\mathbf{r}_i$ as $\mathcal{T}_{\theta_{j\to i}}\mathbf{r}_j = \mathbf{r}_i$. Conversely, if $\theta_{i\to j}$ varies continuously, $\mathbf{r}_i$ will travel smoothly on the manifold $\mathcal{R}$ via $\mathcal{T}_{\theta_{i\to j}}\mathbf{r}_i$. According to the geometry knowledge, $\mathcal{T}_{\theta_{i\to j}}$ corresponds to an affine (a matrix) transformation of the 2-D coordinates of $\mathbf{r}_i$,[1] followed by sampling and interpolation [31], [50]. Therefore, the pose feature $\mathbf{r}_i$ can be characterized by a group of $\{\mathcal{T}_{\theta_{j\to i}}\}_{\forall j\neq i}$, and the entire $\{\mathbf{r}_i\}_{i=1}^{N_c}$ will form a pairwise field that regularizes each component therein. The insight of this strategy is inspired by the principle of relativity, which shows that the characterization of a concept essentially depends on its frame of reference. Accordingly, we do not always need to specify the absolute value of θ_c as supervision, but instead use a set of references to characterize the relative concept. We declare that this type of supervision can contribute to learning the intrinsic representation to answer the above question "what is the pose of a target." If $\theta_{i\to j}$ is known in advance as a prior or computed from pose estimation, it can be considered as deductive supervised information to learn a pose feature extractor $\mathcal{E}$ by minimizing the distance between $\mathcal{T}_{\theta_{i\to j}} \circ \mathcal{E}(\mathbf{x}_i)$ and $\mathcal{E}(\mathbf{x}_j)$. On the contrary, if $\theta_{i\to j}$ is not known, we assume that it can be inferred from $\mathbf{r}_i$ and $\mathbf{r}_j$. For this purpose, the geometric transformation model-based $\mathcal{T}_{\theta_{i\to j}}$ is attached to a pose discrepancy aware network to estimate $\theta_{i\to j}$ as $\mathcal{D}_\varphi(\mathbf{r}_i, \mathbf{r}_j) = \theta_{i\to j}$, and φ contains the parameters of $\mathcal{D}$. Since the geometric component does not need to learn, we call the module semi-parametric [23], which inherits the advantage of both deductive and inductive learning of human learning.

Considering the second challenge, instead of explicitly selecting a suitable $\ell_{\mathcal{R}}$ on the manifold, we will alternatively measure their similarity in another semantically plausible space called the imaginary target subspace. It is motivated by the following consideration. If $\mathcal{R}$ and $\mathcal{F}$ are completely disentangled, we can randomly sample factors from them and create an imaginary target image with the help of $\mathcal{G}_\Theta$. More specifically, let $\{\mathbf{r}_i, \mathbf{f}_i\}$ and $\{\mathbf{r}_j, \mathbf{f}_j\}$ be the pose and identity features of $\mathbf{x}_i$ and $\mathbf{x}_j$, respectively. As mentioned above, $\mathcal{G}_\Theta$ acting as the contextualization head [46] will generate a blurred image $\mathcal{G}_\Theta(\mathbf{r}_j, \mathbf{f}_j)$ defined as the imaginary counterpart of $\mathbf{x}_j$. If it holds, $\mathcal{G}_\Theta(\mathcal{T}_{\theta_{i\to j}}\mathbf{r}_i, \mathbf{f}_j)$ should also produce an analogous result. So we can measure the similarity of $\mathbf{r}_i$ and $\mathbf{r}_j$ in the imaginary target subspace. However, there is a risk that $\mathbf{f}_j$ is not completely decoupled from $\mathbf{r}_j$ and already encodes the pose information. To reduce the risk of information leakage, we use $\mathbf{f}_i$ instead of $\mathbf{f}_j$, so that all the information of $\mathbf{x}_j$ is blind to $\mathcal{G}_\Theta$ during the imagination phase. This strategy consolidates the combinational generation ability and further promotes the identity consistency between $\mathbf{f}_i$ and $\mathbf{f}_j$. Theoretically, any identity vector of the same class can be sampled from $\mathcal{F}$ to replace $\mathbf{f}_j$, but this strategy is not considered in this article. In summary, Ω_r Regularizer, given by

$$\begin{aligned}
&\Omega_r\left(\{\mathbf{r}_j\}_{j=1}^{N_c}\right) \\
&\quad\equiv \begin{cases} \displaystyle\sum_i \sum_j \ell\left(\mathcal{G}_\Theta(\mathcal{T}_{\theta_{i\to j}}\mathbf{r}_i, \mathbf{f}_i), \mathbf{x}_j\right), & \text{if } \mathbf{x}_i, \mathbf{x}_j \in \mathcal{C}^+ \\ \text{const}, & \text{otherwise} \end{cases} \\
&\text{s.t. } \forall i, \quad j,\ \theta_{i\to j} = \mathcal{D}_\varphi(\mathbf{r}_i, \mathbf{r}_j),\ \mathbf{x}_j \leftarrow \mathcal{G}_\Theta(\mathbf{r}_j, \mathbf{f}_j)
\end{aligned} \tag{7}$$

where ℓ is a perception loss function measuring the similarity between an imaginary and a real target.[2] According to the previous analysis, the imaginary target will be blurred with rough orientation and identity rather than details. The common pixelwise metric for image reconstruction, such as mean square error (MSE) or mean average error (MAE), will lead to overregularization. It will encode some additional undesired information in $\mathbf{r}$ and $\mathbf{f}$ for accurate reconstruction. Therefore, a relaxed metric tailored to measure the similarity between an imaginary target and its real counterpart is required.

On one hand, instead of measuring the pixelwise similarity, we alternatively resort to the relaxed statistic similarity. For the true SAR target, the magnitude of pixels will follow a non-Gaussian distribution [27], e.g., Gamma law in (8):

$$p(x|a, b) = \frac{b^a}{\Gamma(a)} x^{a-1} \exp(-bx) \tag{8}$$

where a and b are the distribution parameters, and Γ is the gamma function. Since we expect the imaginary target to keep distribution consistency with its real counterpart, we can choose symmetric Kullback–Leibler (SKL) to measure the distribution discrepancy [51]. More formally, let $p_x(x)$ and $p_{\hat{x}}(x)$ be the magnitude distribution of the pixel in the true and imaginary targets, respectively. SKL distance is defined as

$$\begin{aligned}
&J_D(p_x(x), p_{\hat{x}}(x)) \\
&\quad= \mathrm{KL}[p_x(x)||p_{\hat{x}}(x)] + \mathrm{KL}[p_{\hat{x}}(x)||p_x(x)] \\
&\quad= a_{\hat{x}}(b_x/b_{\hat{x}} - 1) + a_x(b_{\hat{x}}/b_x - 1) \\
&\qquad + (a_x - a_{\hat{x}})\left[\ln(b_{\hat{x}}/b_x) + \Psi(a_x) - \Psi(a_{\hat{x}})\right]
\end{aligned} \tag{9}$$

[1] $\mathbf{r}_i$ will be a set of 2-D feature maps in our situation.

[2] Note that in this article, real and imaginary do not mean real and imaginary parts of a complex number.

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where Ψ is the digamma function. Note from (9) that the SKL distance essentially measures the difference between two pairs of distribution parameters. As a result, they should be estimated from $\hat{\mathbf{x}}$ and $\mathbf{x}$, respectively. In our model, we prefer the following moment estimation $b \leftarrow \text{mean}(\mathbf{x})/\text{var}(\mathbf{x})$ and $a \leftarrow b \cdot \text{mean}(\mathbf{x})$, where $\text{mean}(\mathbf{x})$, $\text{var}(\mathbf{x})$ are the mean value and variance of $\mathbf{x}$, respectively. It can be implemented as a module/layer stacked on top of $\mathcal{G}_\Theta(\mathbf{r}, \mathbf{f})$ allowing backpropagation. It will be a supplementary part of the contextualization head to transform $\mathcal{G}_\Theta(\mathbf{r}, \mathbf{f})$ to the estimated parameters a and b. Then these two values will be used to evaluate the SKL (9).

On the other hand, SKL does not reflect spatial structure information. Any repermutation of the pixels will yield the same SKL. To avoid this ambiguity, we additionally include a second-order statistical metric, namely, the Pearson correlation coefficient, which measures the orientation similarity

$$r(\mathbf{x}, \hat{\mathbf{x}}) = \frac{R\mathbf{x}^T\hat{\mathbf{x}} - \sum_{i=1}^{R} x_i \sum_{i=1}^{R} \hat{x}_i}{\sqrt{R\sum_i x_i^2 - \left(\sum_i x_i\right)^2}\sqrt{R\sum_i \hat{x}_i^2 - \left(\sum_i \hat{x}_i\right)^2}} \tag{10}$$

where R is the dimension of $\mathbf{x}$ or $\hat{\mathbf{x}}$. We finally combine them to form the loss ℓ in (7) as

$$\ell\left(\mathbf{x}_j, \hat{x}_j\right) = J_D\left(p_{\mathbf{x}_j}(x), p_{\hat{x}_j}(x)\right) - r(\mathbf{x}_j, \hat{x}_j). \tag{11}$$

D. Amortized Inference and Overall Architecture

So far, we have developed a contrastive feature disentangling (ConFeDent) model (4), where we elaborate two regularization functions (5) and (7) to inject the identity and pose information into the factors $\mathbf{f}$ and $\mathbf{r}$, respectively. More specifically, Ω_f imposes a discriminative and structural regularization on the pairwise intraclass identity factors. It will not only encode the class-specific information in $\mathbf{f}$ but also promote pairwise similarity. Alternatively, Ω_r involves a semi-parametric geometric transformation model to characterize the relationship between two points on the manifold $\mathcal{R}$ with deductive geometric knowledge. More importantly, we construct the combinational generalization of the imaginary target subspace. Instead of designing a metric on the manifold, we measure the similarity between two pose factors in the imaginary target subspace with an elaborated loss (11), which can further promote feature disentanglement and generalization ability.

Thanks to these strategies, ConFeDent will be more semantically explainable than the other deep-learning-based SAR ATR algorithms. Nevertheless, it is difficult to solve (4) problem with a general algorithm because it is essentially a bilevel optimization. Moreover, feature disentanglement is an inverse problem, which generally requires an iterative optimization process [47], [48]. Inspired by the recently prevalent amortized inference technique [47], we resort to learning an optimizer that directly produces the approximated solutions, yielding an encoder–decoder feature disentanglement architecture. To this end, we first consider the latent features $\mathbf{f}$ and $\mathbf{r}$ and declare that their optimal values can be obtained by a parameterized encoder $\mathcal{E}_\Psi$ as $\mathbf{f}, \mathbf{r} = \mathcal{E}_\Psi(\mathbf{x})$, where Ψ contains the encoder parameters. $\mathcal{E}_\Psi$ can also be viewed as a feature extractor that factorizes the image into the regularized identity and pose factors. Therefore, the final optimization problem for ConFeDent learning is summarized as

$$\begin{aligned} \min_{\Psi,\Theta,\varphi} & \sum_{c=1}^{C} \Omega_f\left(\{\mathbf{f}_i\}_{i=1}^{N_c}\right) + \alpha \sum_{c=1}^{C} \Omega_r\left(\{\mathbf{r}_i\}_{i=1}^{N_c}\right) \\ & + \beta \sum_i \sum_{j\neq i} \ell\left(\mathbf{x}_{i,j}, \mathcal{G}_\Theta(\mathbf{f}_{i,j}, \mathbf{r}_{i,j})\right) \\ \text{s.t. } & \theta_{i\to j} = \mathcal{D}_\varphi\left(\mathbf{r}_i, \mathbf{r}_j\right),\ \mathbf{f}_i, \mathbf{r}_i = \mathcal{E}_\Psi(\mathbf{x}_i) \quad \forall \mathbf{x}_i,\ \mathbf{x}_j \in \mathcal{C}^+ \end{aligned} \tag{12}$$

where the bilevel optimization (4) is relaxed as an unconstrained one, and β is a hyperparameter. This relaxed optimization can be solved efficiently by a standard gradient-based method. For the training phase, several remarks should be emphasized. First, we can sample at most $\mathcal{A}_{N_c}^2$ target pairs from each class to train the model, where $\mathcal{A}$ is the permutation operator. In this respect, ConFeDent will be more suitable for learning with limited training samples due to its improved generalization ability. We encourage the model to use data pairs from cooperative categories at the beginning of the training phase to cover the latent pose manifold. Second, since $\theta_{i\to j}$ and $\theta_{j\to i}$ are reciprocal, it is also suggested to invert $\mathcal{T}_{\theta_{i\to j}}$ to get $\mathcal{G}_\Theta(\mathcal{T}_{\theta_{j\to i}}\mathbf{r}_j, \mathbf{f}_j)$ for additional regularization.

In the deployed testing phase, we remove the projection head $\mathcal{P}$ in Ω_f and the pose contrastive transformer Ω_r. Then ConFeDent becomes a standard fully convolution network (FCN) architecture that can be universally deployed in a general SAR ATR task. For a query sample $\tilde{\mathbf{x}}$, its label vector $\tilde{\mathbf{y}}$ can be efficiently obtained as

$$\tilde{\mathbf{y}} = \mathcal{W}^*(\tilde{\mathbf{f}}), \quad \text{s.t. } \tilde{\mathbf{f}} \leftarrow \mathcal{E}_{\Psi^*}(\tilde{\mathbf{x}}). \tag{13}$$

The overall architecture of ConFeDent for reproduction is illustrated in Fig. 1, and the detailed network information of each module is summarized in Table II.

TABLE II
DETAILED INFORMATION OF EACH MODULE OF CONFEDENT

Module	Architecture	Input	Output
$\mathcal{E}_\Psi$	Conv+MaxPool:K5-S1-C16-ReLU+P2-S2 Conv+MaxPool:K5-S1-C32-ReLU+P2-S2 Conv+MaxPool:K5-S1-C64-ReLU+P2-S2 Dropout Conv:K5-S1-C128-ReLU	$\mathbf{x}$	$\mathbf{f}$
	Conv+MaxPool:K5-S1-C16-ReLU+P2-S2 Conv+MaxPool:K5-S2-C64-ReLU+P2-S2 Dropout Conv:K5-S2-C128-Tanh	$\mathbf{x}$	$\mathbf{r}$
$\mathcal{D}_\varphi$	FC:C60-ReLU FC:C30-ReLU FC:C6	$\mathbf{r}_1$,$\mathbf{r}_2$	$\theta_{1\to 2}$
$\mathcal{G}_\Theta$	ConvT+UpSamp:K5-S1-C64-valid-ReLU+P2-S2 ConvT+UpSamp:K6-S1-C32-valid-ReLU+P2-S2 ConvT+UpSamp:K5-S1-C16-valid-ReLU+P2-S2 ConvT:K5-S1-C1-valid-ReLU	$\mathbf{f}$,$\mathbf{r}$	$\mathcal{G}_\Theta(\mathbf{f}, \mathbf{r})$
$\mathcal{P}$	Identity Mapping	$\mathbf{f}$	$\mathcal{P}(\mathbf{f}) = \mathbf{f}$
$\mathcal{W}$	FC:C-Softmax	$\mathbf{f}$	$\mathbf{y}$

E. Strengthened Version: ConFeDent+

In ConFeDent, the most notable component is the design of the pose contrastive transformer regularizer which aims to

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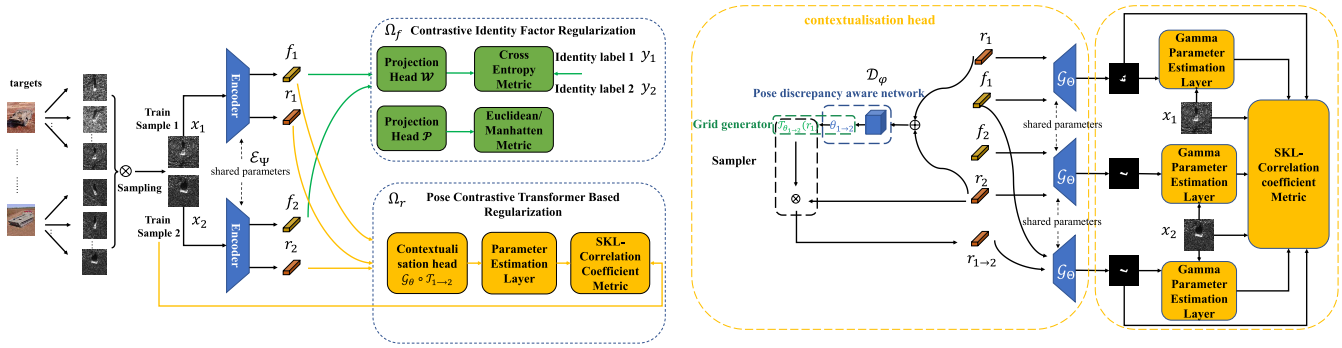


Fig. 1. Framework of the proposed ConFeDent. (a) Overall framework of ConFeDent. (b) Pose contrastive transformer.

warp the pose features $\mathbf{r}_i$ of $\mathbf{x}_i$ to $\mathbf{r}_j$ of $\mathbf{x}_j$ via $\mathcal{T}_{\theta_{i\to j}}$. Instead of directly measuring the similarity in the pose manifold $\mathcal{R}$, we alternatively consider implementing this task in the imaginary target image domain with the help of $\mathcal{G}_\Theta$ as we have the real counterpart of the imaginary target. It follows that the training samples $(\mathbf{x}_i, \mathbf{x}_j)$ have to be restricted as an intraclass pair for self-supervision, which, however, reduces the sufficient use of scarce training data. In addition, knowledge sharing among different categories proceeds implicitly.

The crucial point of the above limitation is the absence of the sampled real NC target corresponding to the imaginary one in the training set. It is not possible to supervise the imaginary NC target image. To overcome this limitation, we propose a strengthened version called ConFeDent+, which can effectively use arbitrary sample pairs for self-training. The motivation of ConFeDent+ is that a robust target imagination should be bidirectional. Specifically, ConFeDent can forward imagine a target object and thus also backtrack from the imaginary counterparts to the original real target. Let $\mathbf{x}_i^1, \mathbf{x}_j^2$ be a pair of class 1 and 2 samples, respectively. Their pose and identity features can be obtained by the well-trained ConFeDent as $\mathbf{r}_i^1, \mathbf{f}_i^1 \leftarrow \mathcal{E}_{\Psi^*}(\mathbf{x}_i^1)$ and $\mathbf{r}_j^2, \mathbf{f}_j^2 \leftarrow \mathcal{E}_{\Psi^*}(\mathbf{x}_j^2)$, respectively. Then ConFeDent can generate two imaginary targets $\hat{\mathbf{x}}_j^1$ and $\hat{\mathbf{x}}_i^2$, which stand for the target in classes 1 and 2 with the approximate pose of $\mathbf{r}_j$ and $\mathbf{r}_i$, respectively. Based on the above motivation, ConFeDent should also backtrack $\mathbf{x}_i^1, \mathbf{x}_j^2$ from $\hat{\mathbf{x}}_j^1$ and $\hat{\mathbf{x}}_i^2$. For this purpose, $\hat{\mathbf{x}}_j^1$ and $\hat{\mathbf{x}}_i^2$ are treated as two latent variables and fed into a next ConFeDent with common parameters in the hope of reconstructing the original $\mathbf{x}_i^1, \mathbf{x}_j^2$. Therefore, ConFeDent+ consists of two ConFeDents with the same parameters in series, as shown in Fig. 2. In the second ConFeDent, we remove the semi-parametric geometric transformation module and pairwise combine the features of $\hat{\mathbf{x}}_j^1$ and $\hat{\mathbf{x}}_i^2$ with $\mathbf{r}_i^1, \mathbf{f}_i^1$, and $\mathbf{r}_j^2, \mathbf{f}_j^2$ into $\mathcal{G}_\Theta$, yielding four outputs. In ConFeDent+, we minimize the MAE loss between $\mathbf{x}_i^1, \mathbf{x}_j^2$ and four outputs instead of the imaginary loss (11). Compared with the architecture in Table II, we slightly modify some building blocks to construct ConFeDent+. More concretely, since ConFeDent+ is a deeper architecture, batch normalization layers are inserted in the pose factor extractor branch in $\mathcal{E}_\Psi$ after each convolution layer. In addition, a "ConvT:K1-S1-C128-valid-ReLU" layer is inserted at the beginning of $\mathcal{G}_\Theta$. The test phase of ConFeDent+ is the same as for ConFeDent.

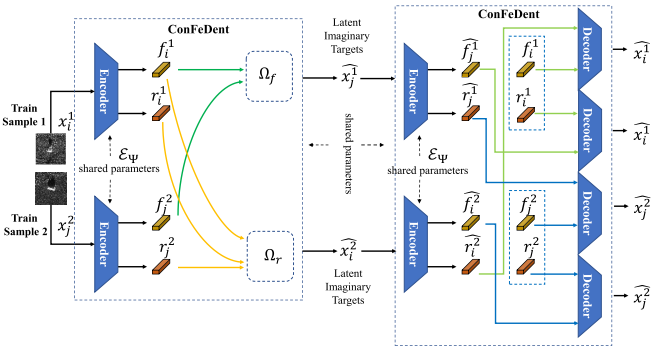


Fig. 2. Framework of the proposed ConFeDent+.

IV. Experiments and Analysis

This section will conduct extensive experiments to evaluate the effectiveness and superiority of ConFeDent and ConFeDent+. First, the experimental dataset MSTAR for PAA ATR and comparison algorithms will be introduced. Second, we design some ablation experiments to validate the effectiveness of the proposed strategies. Finally, we compare ConFeDent and ConFeDent+ with other SAR ATR models to demonstrate their superiority. The overall experiments are performed several times and report the average accuracy.

A. Experimental Settings

1) Dataset Setting: The MSTAR dataset, collected with the Sandia National Laboratory's X-band SAR sensor, is the most widely exploited benchmark for SAR ATR [2]. It contains about ten types of publicly released military ground targets acquired at multiple depression angles and 0°–360° aspect angles with about 5° interval and 0.3 × 0.3 m image resolution, including armored personnel carrier: BMP-2, BRDM-2, BTR-60, and BTR-70; tank: T-62 and T-72; rocket launcher: 2S1; air defense unit: ZSU-234; truck: ZIL-131; bulldozer: D7. Following the common setting [14], the central cropped 88 × 88 magnitude image patches are used for training, validation, and testing. The following experiments will validate the performance of each SAR algorithm under two typical SAR geometric acquisition conditions, i.e., SOC and EOC, each of which incorporates the PAA training setting to increase the difficulty. Specifically, the canonical SOC and EOC-D shown in Tables III and IV consider identical serial

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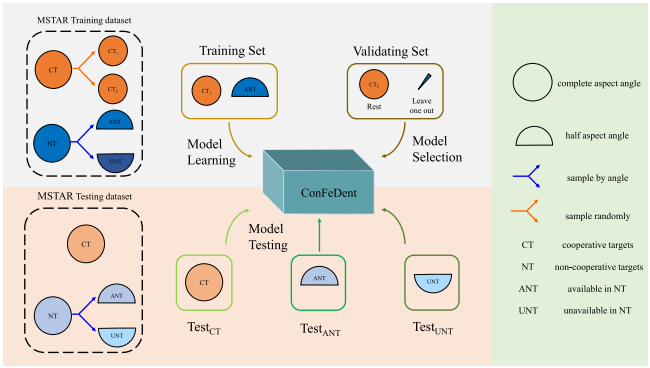


Fig. 3. Illustration of the procedure of dataset splitting.

numbers and configurations with small and large variations in depression angle between the training and test targets, respectively. In addition, two other EOCs of configuration and version variants, namely, EOC-C and EOC-V, shown in Table V, are also evaluated. Given the PAA ATR setting, only a portion of the samples in the training set are selected according to the following operations. First, we randomly select m and n categories as cooperative and NC, respectively, denoted by CmNn. Next, NC targets with 17° depression angle are divided into two groups according to the aspect angle, e.g., $[0, \pi)$ and $[\pi, 2\pi]$, in our experiments. We randomly select only one group of samples called aspect available NC targets (ANT) in the training set for each NC category, leaving one sample in the validation set according to the leave-one-out strategy. The remaining group, called the aspect unavailable NC targets (UNT), is discarded. As a result, about 50% NC targets are used for training. Finally, for cooperative categories, we uniformly select 50% samples with full aspect angle into the training set for class-balance consideration, and the others are added to the validation set. Throughout the experiments, we do not perform any data augmentation procedure for all the algorithms unless specially stated. To clearly demonstrate PAA ATR performance, the test set is correspondingly divided into the following three groups, i.e., targets in the cooperative classes Test_{CT}, NC targets with seen aspect Test_{ANT}, and NC targets with unseen aspect Test_{UNT}. It is worth noting that we should pay more attention to the recognition performance on Test_{UNT} since it requires more generalization ability. Fig. 3 illustrates the above dataset splitting process.

2) Experimental Settings: The parameters in ConFeDent are initially set by default without pretraining. The model is learned using the standard momentum-based stochastic gradient descent, with a learning rate of 0.001 and a momentum rate of 0.9. The batch size is 100. The weight decay regularization with a rate of 0.004 is imposed on the model parameters except φ. We use the early stopping strategy to control the training process by monitoring the recognition accuracy of the validation set. The training process is stopped if the recognition accuracy does not improve within 30 epochs. α and β are 1.5 and 1, respectively, according to the grid search.

B. Ablation Experiments

Without loss of generality, this section will design some specific experiments via ablation to demonstrate the effectiveness of our proposed strategies in ConFeDent under PAA-SOC-C5N5.

TABLE III
TARGET INFORMATION OF SOC

Class	Train(Depression=17°)		Test(Depression=15°)	
	θ_c		θ_c	
	$[0, \pi)$	$[\pi, 2\pi]$	$[0, \pi)$	$[\pi, 2\pi]$
2S1	145	154	137	137
BMP-2	354	344	310	277
BRDM-2	145	153	136	138
BTR-70	108	125	107	89
BTR-60	136	120	103	92
D7	152	147	138	136
T-62	147	152	134	139
T-72	343	348	307	275
ZIL-131	146	153	136	138
ZSU-23-4	145	154	136	138

TABLE IV
TARGET INFORMATION OF EOC-D

Class	Train(Depression=17°)		Test(Depression=30°)	
	θ_c		θ_c	
	$[0, \pi)$	$[\pi, 2\pi]$	$[0, \pi)$	$[\pi, 2\pi]$
2S1	145	154	140	148
BRDM-2	145	153	138	149
T-72	118	114	138	150
ZSU-23-4	145	154	139	149

TABLE V
TARGET INFORMATION OF EOC-C (CONFIGURATION VARIANTS) AND EOC-V (VERSION VARIANTS)

	Class	Serial No.	Depression	No.
Candidate Training Set	T72	132	17°	232
	BMP-2	9563	17°	233
	BRDM-2	E-71	17°	298
	BTR-70	c71	17°	233
EOC-C Testing Set	T72	S7	17°,15°	419
		A32	17°,15°	572
		A62	17°,15°	573
		A63	17°,15°	573
		A64	17°,15°	573
EOC-V Testing Set	T72	812	17°,15°	426
		A04	17°,15°	573
		A05	17°,15°	573
		A07	17°,15°	573
		A10	17°,15°	567
	BMP-2	9566	17°,15°	428
		c21	17°,15°	429

1) Validation of Pose Contrastive Transformer: The main strategy in ConFeDent to improve the generalization ability is the design of Ω, which warps the pose features of one target $\mathbf{x}_i$ to that of the other intraclass target $\mathbf{x}_j$ with the semi-parametric geometric transformation model. To evaluate its ability, we randomly select four pairs of intraclass targets and visualize the imaginary target images $\mathcal{G}_\Theta(\mathbf{r}_{1,2}, \mathbf{f}_{1,2})$, $\mathcal{G}_\Theta(\mathcal{T}_{\theta_{2\to1}}\mathbf{r}_2, \mathbf{f}_2)$, and $\mathcal{G}_\Theta(\mathcal{T}_{\theta_{1\to2}}\mathbf{r}_1, \mathbf{f}_1)$ in Fig. 4. We can see that all the imaginary images are blurred versions of the original inputs with the same orientation. The aspect angles of $\mathcal{G}_\Theta(\mathcal{T}_{\theta_{2\to1}}\mathbf{r}_2, \mathbf{f}_2)$ and $\mathcal{G}_\Theta(\mathcal{T}_{\theta_{1\to2}}\mathbf{r}_1, \mathbf{f}_1)$ are identical to those of $\mathcal{G}_\Theta(\mathbf{r}_1, \mathbf{f}_1)$ and $\mathcal{G}_\Theta(\mathbf{r}_2, \mathbf{f}_2)$, although the aspect angle discrepancy between $\mathbf{x}_1$ and $\mathbf{x}_2$ is large. Therefore, these results can demonstrate the effectiveness of the proposed model for

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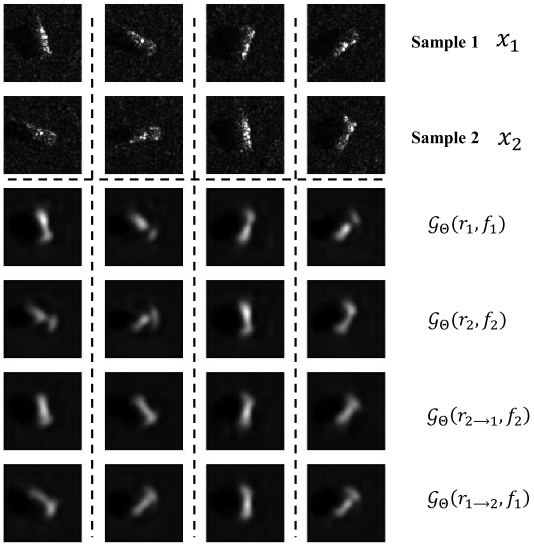


Fig. 4. Illustration of intraclass cross-imagination ability of ConFeDent.

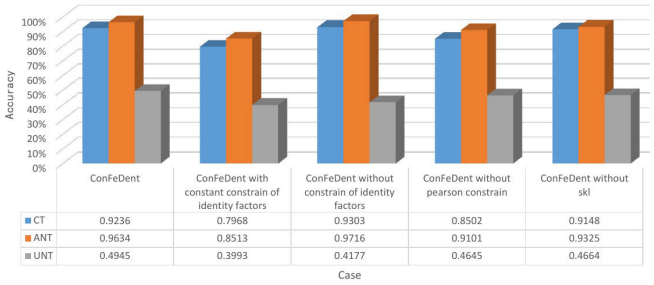


Fig. 5. Classification accuracy of ablation experiments in SOC-C5N5.

disentangling the pose factors among intraclass samples and cross-transformation.

2) Evaluating the Strategies in Ω_f and Ω_r: We introduced a dynamic hyperparameter μ in Ω_f to avoid over-penalization and used a curriculum schedule to incrementally increase μ. In Ω_r, we elaborate a new similarity metric (11) that combines the first-order SKL and second-order Pearson correlation coefficient functions. It measures the pose similarity between a blurred imaginary SAR target and its real counterpart. To verify the effectiveness of the above strategies, we conduct several experiments to evaluate the impact on classification accuracy and visual results under the following conditions: 1) the standard ConFeDent; 2) specifying $\mu \neq 0$ via cross-validation; 3) $\mu = 0$; 4) removing Pearson correlation from (11); and 5) removing SKL from (11). The corresponding results are summarized in Figs. 5 and 6. Let us observe the classification accuracies in Fig. 5. The accuracy of Case 2 is always lower than Case 1 and Case 3, which is caused by the over-penalization of the pairwise potential function. Comparing Case 1 with Case 3, although Case 3 can achieve slightly higher accuracy on Test_{CT} and Test_{ANT}, it performs much worse on Test_{UNT}. Comparing the results of Case 1, Case 4, and Case 5, we find that both the Pearson correlation coefficient and the SKL distance help improve the classification accuracy. These results show that ATR will benefit from feature disentanglement due to the improved generalization ability. Consider the visual results shown in Fig. 6. Comparing Fig. 6(a) with Fig. 6(b) and (c), there is no obvious variation in the resulting images. Nevertheless, the cross-transformations of the first, second, and fourth target pairs in Fig. 6(c) by removing the pairwise term are failed ones, which implies its influence on pose factor disentanglement. Alternatively, in Fig. 6(d), removing the Pearson correlation coefficient will generate more blurred images which cannot recognize any orientation information. Finally, in Fig. 6(e) and (f), removing the SKL distance will generate darker images that cannot obtain any useful information. For better illustration, we normalize the pixel values to the range [0, 1] for contrast enhancement. We can see that the imaginary targets are more compact and complete, but the corresponding pixel values are low. It means that the statistical distribution characteristics are different from the original image. In summary, SKL and Pearson correlation will play distinct roles in feature regularization.

C. Algorithms Comparison on SAR ATR

In this section, we will compare the performance of ConFeDent and other algorithms on the SAR NC-ATR task under different conditions. First, we compare the general SAR ATR performance under SOC with reduced training samples. Next, we combine the PAA condition with SOC, EOC-D, EOC-C, and EOC-V to evaluate ATR performance. Finally, the algorithms are compared on much more complicated PAA conditions where the available range of aspect angles of NC is only $[0, 2\pi/3]$ and $[0, \pi/2]$, respectively.

1) Comparison Results Under SOC: The first comparison experiment is performed on SOC, whose training and test samples are the original ones in Table III without considering the PAA. To demonstrate the effectiveness and superiority of ConFeDent and ConFeDent+ for general SAR-ATR, we compare the results with those of the following typical models: A-ConvNet [14], RotANet [21], K-nearest neighbor (K-NN), and sparse representation classifier (SRC) [9]. A-ConvNet achieves state-of-the-art accuracy with a standard FCN architecture and is therefore treated as a baseline SAR ATR algorithm. We also use the data augmentation strategy to manually generate the pseudotargets with full aspect angles for NC categories. They are used to train A-ConvNet, resulting in a variant denoted as A-ConvNet*. In addition, several mainstream deep network models for optical image classification are also compared by fine-tuning the pretrained models, including ResNet50 [52] and VGG [53]. In the experiment, we vary the training sample rate from 10% to 50% with an interval of 10%. The comparative recognition accuracy is plotted in Fig. 7. As can be seen from the figure, ConFeDent and ConFeDent+ can outperform other compared algorithms in most cases. More importantly, the accuracy curves of ConFeDent+ are flatter than others, indicating its robustness in reducing the number of training samples.

2) Recognition Results Under PAA-SOC: The following experiments will validate several situations that consider training with different numbers of NC categories, including C5N5, C1N9, and C0N10, according to the setting

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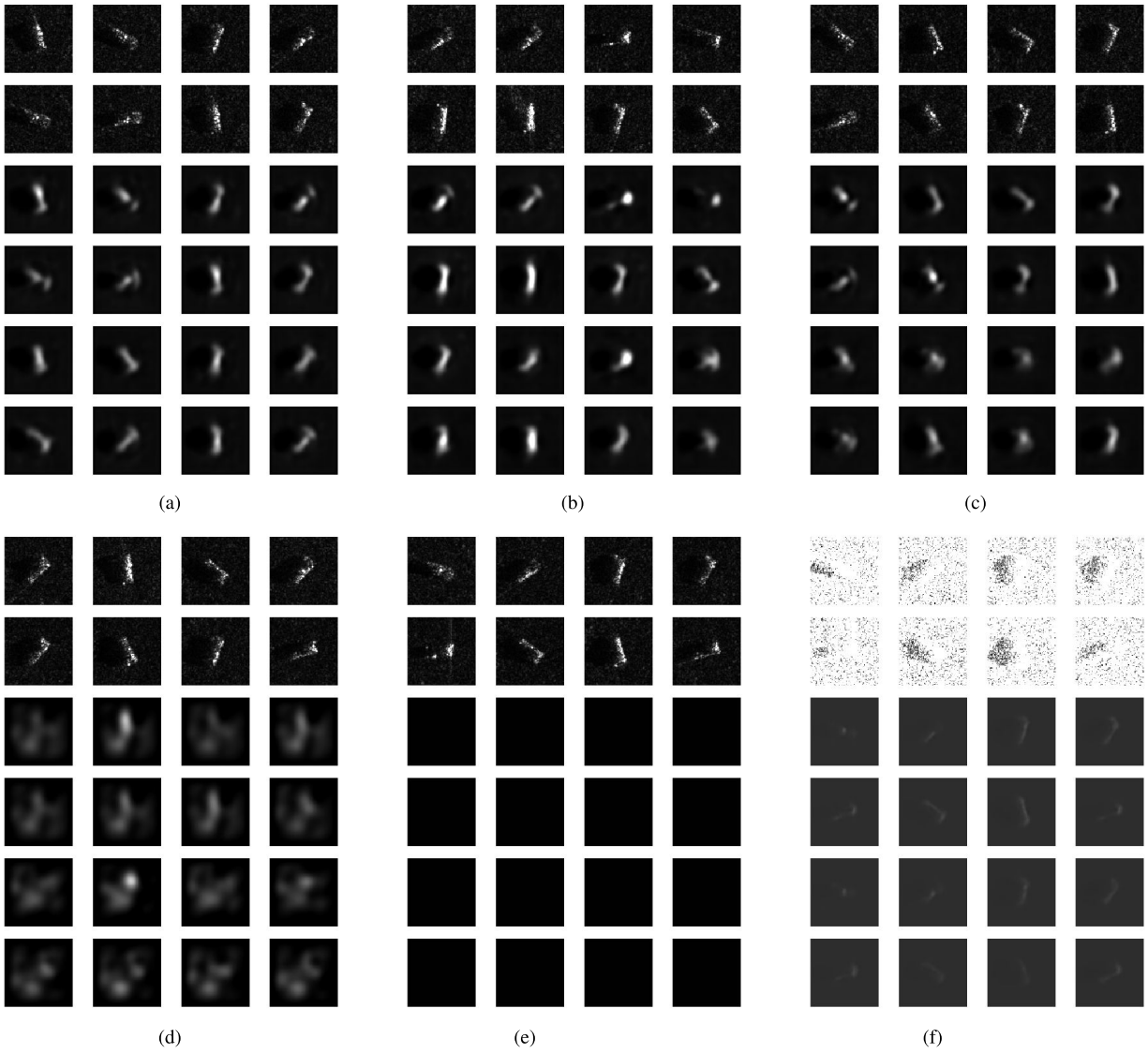


Fig. 6. Ablation experiments on different strategies in Ω_f and Ω_r. IDR is an abbreviation for image display range. (a) Case 1. (b) Case 2. (c) Case 3. (d) Case 4. (e) Case 5 [IDR(0,40)]. (f) Case 5 [IDR(0,1)].

TABLE VI
RECOGNITION AVERAGE ACCURACY OF DIFFERENT ALGORITHMS UNDER PAA-SOC

Methods	C5N5			C1N9			C0N10	
	CT	ANT	UNT	CT	ANT	UNT	ANT	UNT
SRC	91.06%	95.99%	31.88%	94.83%	96.31%	33.43%	96.50%	36.11%
A-ConvNet	89.77%	92.83%	23.67%	***96.13%***	***97.15%***	31.07%	***97.29%***	33.98%
A-ConvNet*	91.49%	96.05%	28.29%	90.95%	94.72%	27.44%	96.04%	29.43%
A-ConvNet+STN	92.21%	***97.39%***	38.92%	93.46%	95.79%	32.76%	95.70%	33.77%
InceptionV3	77.97%	83.85%	38.62%	80.08%	82.23%	41.02%	84.90%	39.21%
ResNet50	64.38%	71.85%	32.33%	65.85%	75.40%	36.51%	76.34%	37.49%
VGG16	87.94%	93.24%	39.39%	82.10%	87.52%	38.49%	95.11%	37.80%
ConFeDent	***92.36*** %	96.34%	***49.45***%	90.92%	94.00%	***43.78***%	96.96%	***43.86***%
ConFeDent+	**93.84** %	**97.86**%	**56.73**%	**97.27**%	**97.19**%	**49.10**%	**97.86**%	**55.87**%

in Section IV-A. In addition to the aforementioned algorithms, STN is a plug-and-play module for generating spatial transformation-invariant features [31]. We add an STN module to A-ConvNet, resulting in A-ConvNet + STN for comparison. The detailed classification results of all the algorithms are summarized in Table VI, where the top 2 results are marked with **bold** and ***italic*** fonts, respectively. Checking the results, most algorithms can achieve satisfactory accuracy on Test_{CT} and Test_{ANT} in three situations. In particular, ConFeDent+ achieves top-1 accuracy in all the cases. For Test_{UNT} in three cases, ConFeDent+ can improve the accuracy of rank-2 ConFeDent by 5%–12%, and ConFeDent will further improve

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TABLE VII
RECOGNITION AVERAGE ACCURACY OF DIFFERENT ALGORITHMS UNDER PAA-EOC-D

Methods	C3N1			C2N2			C1N3			C0N4	
	CT	ANT	UNT	CT	ANT	UNT	CT	ANT	UNT	ANT	UNT
A-ConvNet	***79.58%***	78.59%	29.22%	***80.54%***	78.24%	35.29%	82.20%	***84.75%***	32.14%	84.94%	40.42%
A-ConvNet*	76.32%	***89.05%***	55.39%	75.71%	***88.13%***	46.31%	76.96%	83.17%	42.04%	81.57%	38.88%
A-ConvNet+STN	76.71%	83.56%	51.18%	**80.94%**	78.67%	44.98%	73.81%	71.31%	44.37%	**85.77%**	***53.97%***
InceptionV3	59.16%	58.70%	47.21%	58.38%	58.86%	44.51%	54.09%	60.72%	49.20%	60.99%	48.73%
ResNet50	41.96%	38.93%	29.91%	36.37%	27.79%	25.87%	37.75%	32.28%	29.66%	28.07%	25.53%
VGG16	69.25%	80.44%	48.21%	64.10%	73.07%	50.91%	66.22%	79.22%	49.22%	61.51%	46.87%
ConFeDent	75.42%	88.28%	***73.28%***	74.98%	84.10%	***58.66%***	***82.58%***	76.83%	***49.65%***	68.08%	53.36%
ConFeDent+	***76.83%***	**94.44%**	**79.45%**	74.83%	**91.40%**	**68.34%**	**87.93%**	**88.68%**	**54.88%**	***85.16%***	**54.87%**

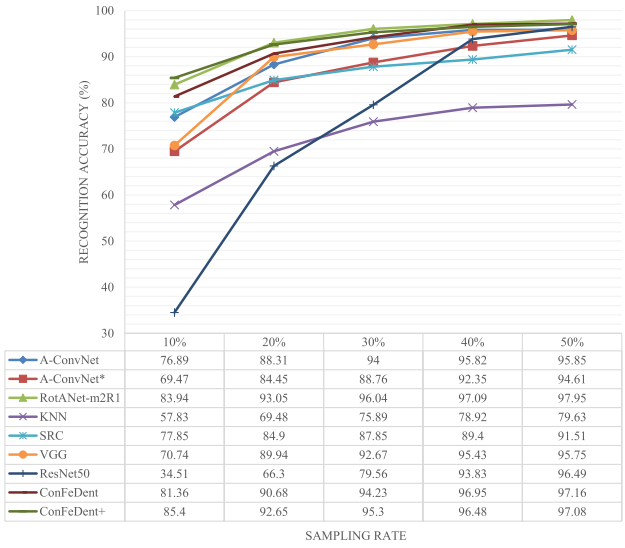


Fig. 7. Comparison recognition accuracy under SOC.

the accuracy in Test_{UNT} by 10% on average than others. This phenomenon clearly demonstrates the superiority of ConFeDent and ConFeDent+ in addressing PAA ATR. Comparing the accuracy of ConFeDent and A-ConvNet on three Test_{UNT}, we can see that ConFeDent can achieve 25.78%, 12.71%, and 9.88% improvement. It can demonstrate that ConFeDent can benefit from the increased amount of cooperative samples. The result of A-ConvNet* trained with the manually generated pseudo targets is lower than that of A-ConvNet. It confirms that the data augmentation trick used in optical image classification may no longer be appropriate for SAR ATR. From the accuracy of A-ConvNet + STN, we can conclude that the STN module can slightly improve the performance of A-ConvNet in most cases. However, due to different mechanisms and motivations, the generalization ability of STN on Test_{UNT} is still weaker than our proposed ConFeDent, and it is less efficient for STN to tackle the o.o.d. classification task. For three popular pretrained deep learning architectures, VGG can achieve better performance than the other two. In the extreme case of C0N10, where there is no cooperative category, we assume that the aspect angles of entire NC categories can sufficiently cover the range of $[0, 2\pi]$. According to the result, ConFeDent and ConFeDent+ can also outperform the other algorithms, which further confirms the superiority of knowledge sharing.

TABLE VIII
RECOGNITION AVERAGE ACCURACY OF DIFFERENT ALGORITHMS UNDER PAA-EOC-C AND PAA-EOC-V

Method	PAA-EOC-C		PAA-EOC-V			
	C4N0	C3N1	C4N0	C3N1		C2N2
	CT	NT	CT	CT	NT	NT
A-ConvNet	83.80%	53.70%	82.36%	82.90%	53.73%	59.01%
ResNet50	85.36%	68.13%	72.62%	82.51%	61.29%	***59.64%***
ConFeDent	***93.94%***	***74.87%***	***91.73%***	***87.71%***	***62.27%***	59.44%
ConFeDent+	**95.55%**	**75.17%**	**93.57%**	**90.99%**	**64.37%**	**63.61%**

3) Comparison Results of PAA-EOC-D: Next, we will perform the PAA-EOC-D comparison experiments. In this combined condition, the training and test samples are acquired at a larger depression angle (17° versus 30°), resulting in a more challenging o.o.d. classification task than SOC. Since the number of all target categories in EOC-D is smaller than in PAA-SOC, we can use $\sum_{n=1}^{4} C_4^n$ combinations of the NC class from C0N4 to C3N1 in total. The comparison results are summarized in Table VII. During the experiments, we especially observed a new challenging phenomenon that the performance of different NC combinations varies significantly. The recognition accuracy for some combinations is surprisingly unstable for all the algorithms, especially for ConFeDent and ConFeDent+. We suspect that the reason is that the four target types in EOC-D are very different. Specifically, 2S1 is a carriage-mounted howitzer, BRDM-2 is an amphibious armored reconnaissance vehicle, T72 is a main battle tank, and ZSU-23-4 is an anti-aircraft vehicle. According to their appearance, T72 and 2S1 are very different from BRDM-2 and ZSU-23-4. Intuitively, their information will be more difficult to share, in which case the performance of ConFeDent and ConFeDent+ will be degraded. Then the overall average accuracy will be lowered. Nevertheless, it can also be observed from the results in Table VII that ConFeDent+ achieves the best performance in most cases, and ConFeDent can also outperform other compared algorithms on UNT. Under the extreme condition C0N4, our proposed methods can also achieve better performance. Similar to the results of previous experiments in the PAA-SOC condition, the STN module can improve the recognition performance of A-ConvNet on Test_{UNT}. It proves that our proposed method can effectively use the additional information transferred from CT.

4) Comparison Results of PAA-EOC-C and PAA-EOC-V: In these cases, the categories between training and testing

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TABLE IX
COMPARISON ACCURACY IN SMALLER RANGE OF ASPECT ANGLES ON PAA-SOC C5N5

Methods	$[0, 2\pi/3]$			$[0, \pi/2]$		
	CT	ANT	UNT	CT	ANT	UNT
AconvNet	82.19%	97.21%	12.74%	78.82%	94.34%	22.85%
AconvNet*	87.86%	95.27%	16.09%	86.59%	96.69%	14.04%
AconvNet+STN	89.99%	97.53%	***22.07%***	87.93%	95.69%	18.10%
InceptionV3	61.73%	78.79%	18.05%	78.96%	84.96%	28.02%
ResNet50	34.68%	48.99%	18.66%	64.45%	78.56%	24.58%
VGG16	86.69%	93.73%	14.53%	90.18%	95.11%	26.52%
ConFeDent	***91.01%***	***97.63%***	17.67%	***92.50%***	**98.06%**	***29.41%***
ConFeDent+	**92.10%**	**97.84%**	**25.63%**	**93.63%**	***97.67%***	**36.97%**

are disjoint, which significantly increases the challenge. Note from Table V that the training categories are T72-132, BMP2-9563, BRDM2-E71, and BTR-70, but the test samples are different configuration variants of T72 in EOC-C and different versions of T72 and BMP2 in EOC-V. Considering the PAA setting, only T72-132 is treated as NC or not in EOC-C, resulting in two test scenarios of C4N0 and C3N1. Then, T72-132 or BMP2-9563 or both will be treated as the NC in EOC-V, resulting in three test scenarios of C4N0, C3N1, and C2N2. Table VIII shows the comparison results with A-ConvNet and ResNet50, where the average accuracy of CT and NC detection is reported. According to the results, our two proposed models can outperform the other two algorithms in most cases. In particular, for normal EOC without PAA setting, namely, C4N0, ConFeDent and ConFeDent+ can achieve higher accuracy than the compared two typical algorithms in both the conditions. The results show that ConFeDent and ConFeDent+ can learn more generalized features in the case of configuration and version variants and can maintain the similarity among intraclass samples with slight configuration and version changes. It is mainly because the pairwise regularization term Ω_f encourages the intraclass identity features to be close.

5) Comparison Results of Smaller Range PAA: Finally, we will conduct comparative experiments under more severe PAA conditions by considering smaller ranges of aspect angles. For this purpose, we test two situations where the range of aspect angles of the NC for training is $\pi/2$ and $2\pi/3$. Accordingly, we can divide the original training samples in Table III into four and three parts corresponding to the situation of $\pi/2$ and $2\pi/3$, respectively. We randomly select one part of the samples for the actual training for each NC. Through this operation, the corresponding training samples will be approximately $1/4$ and $1/3$ of the original ones. As a typical example, the algorithms are tested again on the PAA-SOC of the C5N5 dataset, and Table IX summarizes the comparison results. Although in the $\pi/2$ case the accuracy of ConFeDent on Test_{UNT} is not the highest, it is undoubtedly the best-performing method under all the conditions combined. In the case of $2\pi/3$, ConFeDent achieves the best classification results under all the conditions. A-ConvNet* performs worse than its original model. It can be seen from the results that manual rotation in the CT condition improves the classification effect by increasing the diversity of the samples to improve the generalization of the method in this condition. However, this operation is not equipped to deal with the recognition problem under o.o.d.. The STN module improves the classification performance on Test_{UNT} while reducing the classification performance on the other two conditions, and while this has some application value, it is a compromise. While both ResNet50 and InceptionV3 perform well on Test_{UNT}, similar to the STN module, they lose out on the other conditions, especially ResNet50.

V. CONCLUSION

This article proposes a novel contrastive feature disentangling model called ConFeDent and a strengthened version ConFeDent+ to induce features with improved generalization and interpretation ability for SAR ATR. The proposed models develop a novel idea of learning to disentangle the pose and identity features by comparing the semantic relations between two arbitrary SAR targets instead of treating them independently. A semi-parametric geometric transformation model and a second-order energy model are designed for contrastive regularization, where in particular we exploit the deduction-based geometric knowledge as supervision to teach the model to learn the semantic concept of aspect angle. Then, the induced pose and identity feature subspaces from all the training samples will contain complete aspect angle and class-specific information, respectively, enhancing the combinational generalization ability. By taking the best of both worlds, it inherits the benefit of both deductive and inductive learning, analogous to human learning, making the features more interpretable. Compared with the other SAR ATR algorithms, the proposed models can achieve better performance under the general SOC. In particular, under the more complicated PAA-SOC, PAA-EOC-D, PAA-EOC-C, and PAA-EOC-V scenarios, our models can outperform the compared algorithms by a large margin.

Several statements need to be clarified. First, it should be noted that changes in the angle of depression and the resulting angle of incidence on the ground can cause nonlinear changes in regions with topography. It is assumed that the results presented in this study were tested in areas with flat topography, where it is easier to relate changes in depression angle to backscattering behavior. However, it should be acknowledged that the proposed strategy may need to be modified if the target is located in an area with sloping topography. In addition, in the training process of ConFeDent and ConFeDent+, cooperative samples are treated with the same importance as NC samples. Future research will explore the significant contributions of these cooperative training samples through a meta-learning paradigm.

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